



PIXEL

A Competent Book on Computer Education with AI + ML



Shashank Joshi

(M. Tech.)

1



I am



I choose **JPEN Books** because ...

- **JPEN Books** are based on an Interactive Technique that makes students Highly Motivated to gain confidence in their own Abilities.
- **JPEN Books** use conceptual learning in a step-by-step method.
- **QR Codes** are provided for every chapter to unlock the very Fine Digital Content.



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Preface

Today, we are living in the Digital technology world so we cannot ignore the importance of computer. The impact of technology in our modern-day life is such that it has become more of a necessity for us rather than a facility. It is now an all-invasive component of our life, be it an education, occupation or our common life. It is an important tool that adds massive value to the process of teaching and learning.

Our series **Pixel** is strictly based on the guidelines given by **National Education Policy (NEP)** and **National Curriculum Framework (NCF)**. Young children need to be given the tools and techniques to learn, master and integrate the use of computer in their learning life from the beginning.

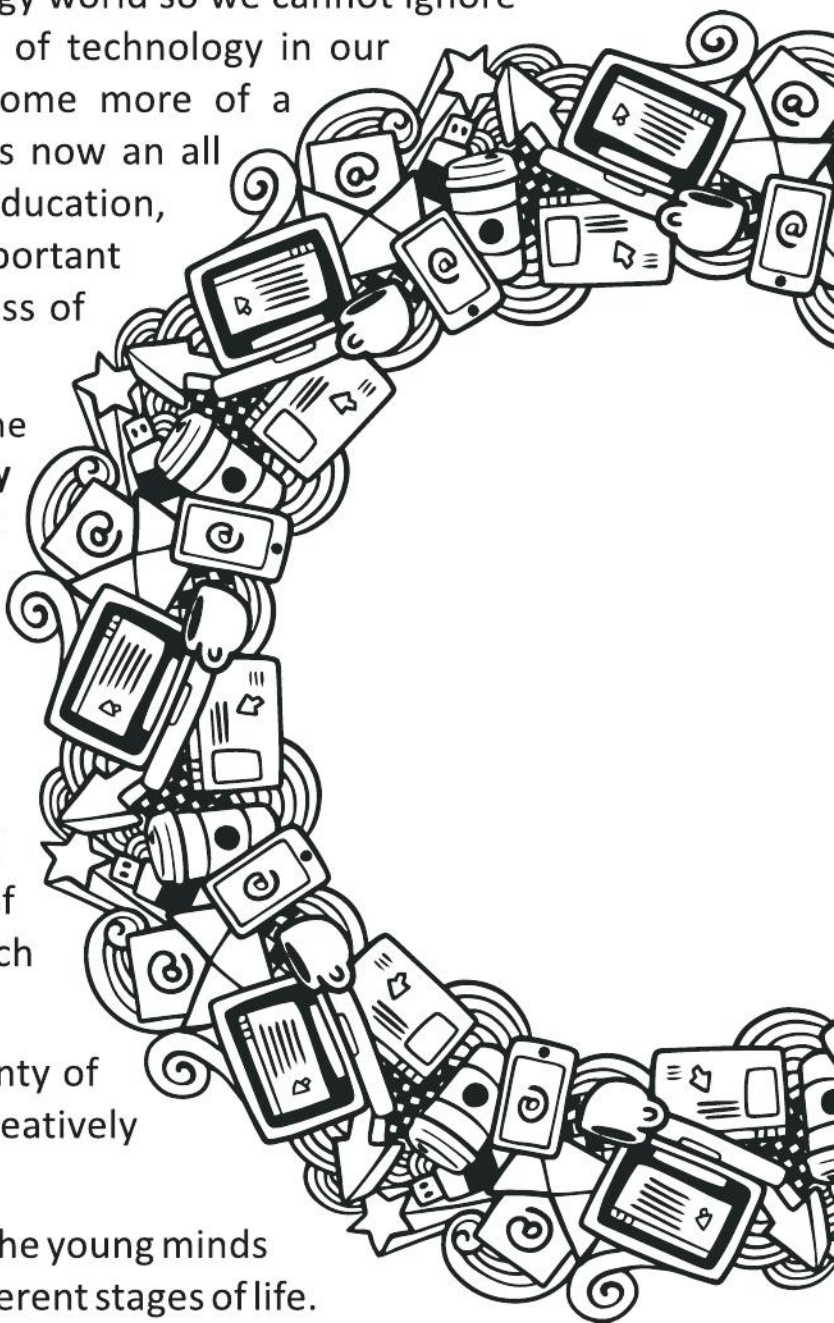
While creating this series, we took special care of few things that happen in the world of Digital technology, latest updates of software versions, and an all-new approach to bridging the digital learning.

This is an activity-oriented series with plenty of screen captures, colourful illustrations, creatively designed exercises and activities.

We sincerely hope that our books will help the young minds to equip them with the required skills at different stages of life.

Any suggestions for the improvement of these books will be highly appreciated.

—Author and Publishers



INCORPORATES NEP 2020 AND NCF 2022

Created with a new dimension of joyful learning along with completely mapped parameters of National Education Policy, 2020

01

CRITICAL THINKING

Strengthening decision-making and innovation

02

ART INTEGRATION

Promoting art-based learning

03

CREATIVE LEARNING

Thinking out of the box

04

SUBJECT ENRICHMENT

Aimed at understanding and skill developing in specific subject

05

INTER DISCIPLINARY

Connects knowledge across the curriculum for holistic development

06

COMPUTATIONAL THINKING

Enables to help solve problems and make thinking more engaging.

08

LIFE SKILLS AND VALUES

Inclined towards empowering moral values in learners

07

EXPERIMENTAL LEARNING

Encourages learning by doing through hands-on activities

09

COMMUNICATION

Emphasis on soft skills and power of language



SYNOPSIS



Outline includes the topics to be learnt in the chapter.



Outlines

- Computer : An Electronic Machine
- Features of a Computer
- Types of a Computer

Info Zone include some facts related to chapter to enhance the knowledge of the students.



The first version of **MS Paint** was introduced with the first version of windows in 1985.

Smart Tip provides keyboard shortcuts for menu commands, to help users save time while performing routine operations.



Smart Tip

You can also open **Paint** by typing Paint in the **Search Box** next to the **Start** button. Then, click on the **Paint** button.

Computer Etiquette presents computer ethics and manners to be followed while working on a computer.



COMPUTER ETIQUETTES

While working on a computer, keep your neck relaxed. Practise this exercise to avoid any strain on your neck.

Let's Implement contains carefully graded exercises to test the knowledge of concepts learnt in a chapter.



Let's Implement.

A Tick (✓) the correct option.

- A monitor is also called :
(a) LCU ☐ (b) VDU ☐
- Which part of a computer is called the brain of a computer?
(a) CPU ☐ (b) VDU ☐
- It is used to give the output of our work on paper.
(a) Speakers ☐ (b) Printer ☐

Refreshment Corner includes interesting exercises for enhancing the skills with fun.



Refreshment Corner

A Colour the alphabet keys that spell your name with yellow.



Practical Session for practical hands-on exercises to strong then the concepts learned by doing.



Practical Session

Experiential Learning

- Visit your computer lab and open Paint. Draw and colour the following.



Just Have a Discussion stimulates a small discussion for the topics being covered in the chapter which in turn improves the communication and logical skills.

Just Have a Discussion

Discuss the different components of ScratchJr in class in detail.

Communication

Racking Your Brain includes the life skills and values questions.

RACKING YOUR BRAIN

Life Skills and Values

Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

Teacher's Note provides important information and suggestions on creative approaches to a chapter or a topic.



TEACHER'S NOTE

- Give enough exercises to students so that they get practice of using the tools of Paint as well as using the mouse.

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COMPUTER



Outlines

- Machines
- Function of Machines



Machines

This is a Television.
It is used to watch the programs.

This is a Mobile Phone.
It is used to make and receive a call.



This is an Air Conditioner.
It is used to keep our home cold.

This is a CD-Player.
It is used to play the music.





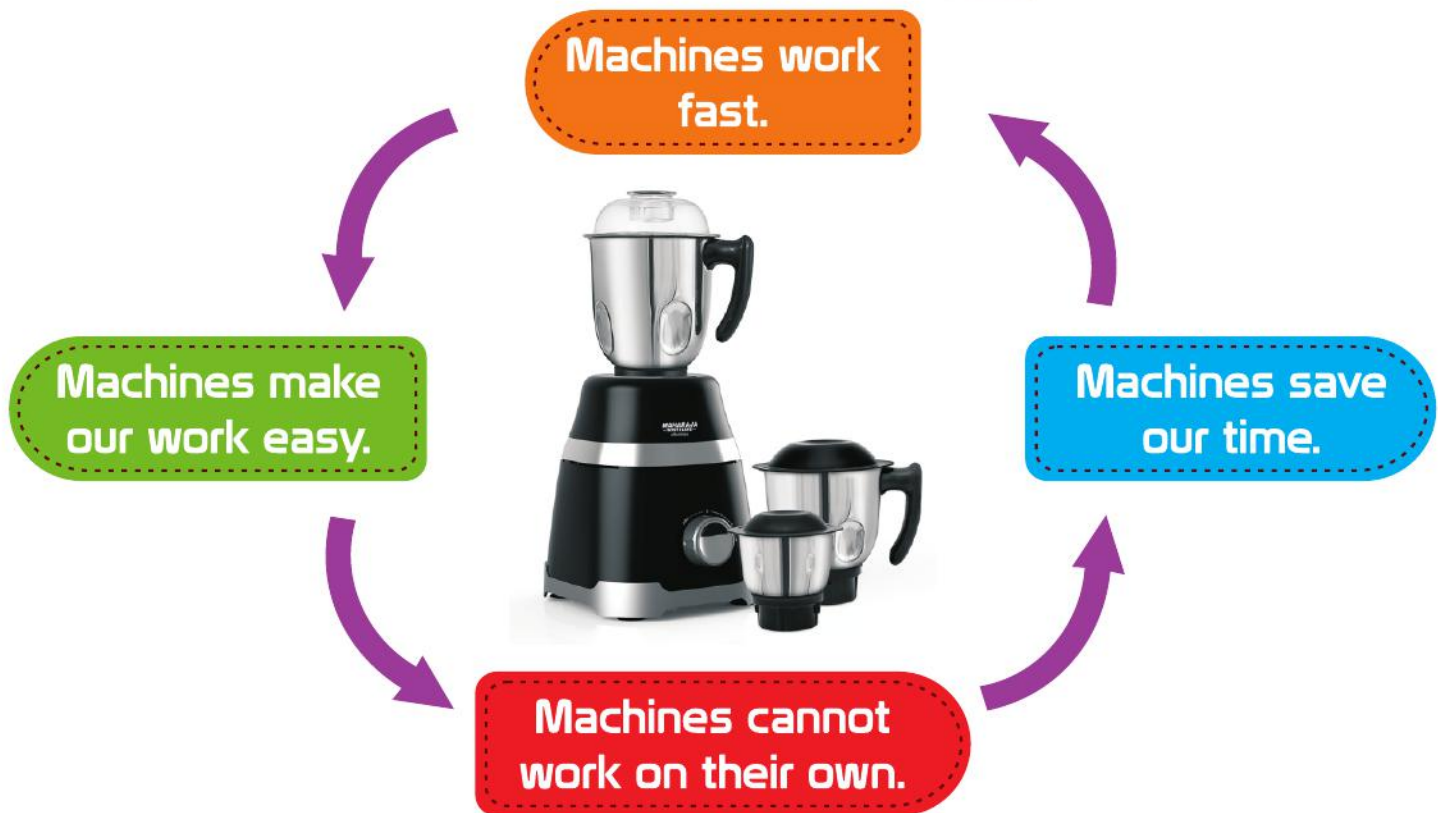
This is a Washing Machine. It is used to wash our cloths.

This is a Refrigerator. It is used to keep our food cool and fresh.



Machines make work easier by changing the way we use force and movement.

Functions of Machines



Most of the machines need electricity to work. In the same way computer is also a machine.



So, we can say that :

- ◆ Computer is also a Machine which works on electricity.
- ◆ Computer can do many things.
- ◆ It works very fast without making any mistakes.
- ◆ We can play games on a computer.
- ◆ We can do calculations and solve problems on a computer.
- ◆ But a computer cannot work on its own it makes our instruction to do work.



The Computer memory is called RAM.



Let's Implement.

A Tick (✓) the correct option.

1. Machines make our work :

(a) Easy



(b) Tough



2. Machines work very :

(a) Fast



(b) Slow



3. Machines saves our :

(a) Time



(b) Money



4. A computer runs on :

(a) Fuel



(b) Electricity



B Fill in the blanks.

1. Computer is a _____.

2. _____ is used to keep the food cool.

3. Machines make our work _____.

4. Computer work very _____.

5. Computer needs our _____ to do work.

C Match the following.

1. To watch cartoons

(a)



(b)



2. To make juice

(c)



(d)



3. To listen songs

4. To talk others

(e)



5. To wash cloths

D

-



-



RACKING YOUR BRAIN

Life Skills and Values

Your computer in the school lab is not working properly, what will you do?



TEACHER'S NOTE

Explain to students about some more machines.



USES OF A COMPUTER

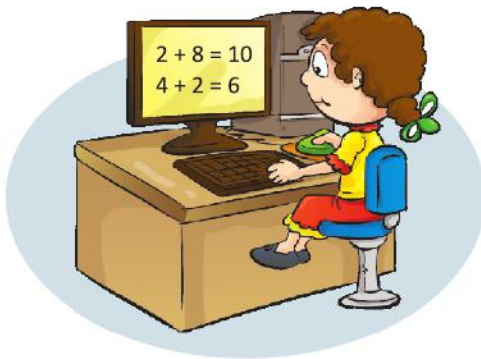


Outlines

- Various Uses of a Computer
- Places Where Computers are Used

Various Uses of a Computer

Various uses of a computer are as follows:



Computer helps in solving the sums.

Computer helps in playing the games.



Computer helps in storing the information.





Computer can do many things at one time.

Computer helps in watching cartoons and movies.

Computer helps in making drawings.



Computer helps in listening music and songs.



Computer helps in writing the letters.



Places Where Computers are Used

Let us learn about various places where computers are used:

At Home

- To play games.
- To watch cartoon and movies.



Uses of a Computer



At the School

- To teach students.
- To make report cards.
- To keep records of students and teachers.

At the Office

- To keep accounts.
- To type the letters.
- To use internet.



At the Shop

- To make bills.
- To keep records of the items.



You can also send a e-mail by computer.



At the Library

- To store a list of books and their locations.
- To issue the books.

At the Bank

- To store a data of people and their accounts.
- To give the money from ATM.



At the Railway Station

- To book tickets for train.
- To maintain train timings.

At the Hospital

- To keep records of patients.
- To treat patients and help in operations.



Let's Implement.

A Tick (✓) the correct option.

1. Computer is used to keep records of students and teachers:

(a) at the hospital

☐

(b) at the school

☐

2. Computer is used to keep records of patients:

(a) at the hospital

☐

(b) at the bank

☐

3. Computer is used to store a data of people and their accounts:

(a) at the office



(b) at the bank



4. Computer is used to store a list of books and their locations:

(a) at the library



(b) at the office



B Fill in the blanks.

1. Computer helps in solving the ___ U ___.

2. Computer helps in playing the G ___ E ___.

3. Computer helps in information ___ T ___ R ___ N ___.

4. Computer helps in watching ___ A ___ T ___ N ___ and
___ O ___ I ___.

C Write True or False.

1. Computer helps in making drawings.

2. Computer does not solve the sums.

3. Computer helps in listening music and songs.

4. Computer does not help in writing the letters.

D Match the names with the pictures.

1.

Hospital

2.

Railway Station

3.

School

4.

Bank

(a)



(b)



(c)



(d)





Refreshment Corner



- Find the hidden words in the crossword. One is done for you.

School

Bank

Shop

Hospital

Station

Airport

Office

P	Y	T	R	D	X	K	H	B
S	T	A	T	I	O	N	O	A
H	N	W	S	M	I	H	S	N
S	C	H	O	O	L	N	P	K
O	U	S	G	J	Z	X	I	C
O	F	F	I	C	E	B	T	T
A	X	S	H	O	P	V	A	A
A	I	R	P	O	R	T	L	L

RACKING YOUR BRAIN

Life Skills and Values

Have you noticed computers being used at other places? What were they being used for?

Just Have a Discussion

Communication

Discuss why it is important to know how to work on a computer in today's world.



TEACHER'S NOTE

Ask the students about different places where they have seen the use of computers.



PARTS OF A COMPUTER



Outlines

- Monitor
- CPU
- Keyboard
- Mouse
- Printer
- Speakers

Parts of a Computer



This is a computer. It has many parts. Computer is a machine used to play games, calculate numbers, draw pictures and do many more things.

Monitor



This is a Monitor. It looks like a television. It displays all the work which we do on computers. The display on monitor is called a soft copy.



Monitor is also called **V.D.U.** (Visual Display Unit).

CPU

This is a CPU. The full form of CPU is Central Processing Unit. All the work we do on computer, is controlled by a CPU.



The CPU is called a brain of the computer. It is the most important part of a computer system.

Keyboard



This is a Keyboard. It contains many keys like- alphabet keys, number keys, symbol keys, etc.

Keyboard is used to give instructions to the computer. Nowadays wireless keyboard is also in use.



Mouse



This is a Mouse. It is used to give instructions to the computer by pointing and clicking. Nowadays wireless keyboard is also in use.



Nowadays, wireless mouse is also in use.



Printer



This is a Printer. It is used to take the printout of our work.

A printout from the printer is called **hard copy**.



Speakers



These are Speakers. They are used to listen to music and songs.





Let's Implement.

A Tick (✓) the correct option.

1. A monitor is also called:

(a) LCU

☐

(b) VDU

☐

2. Which part of a computer is called the brain of a computer?

(a) CPU

☐

(b) VDU

☐

3. It is used to take the printout of our work.

(a) Speakers

☐

(b) Printer

☐

B Fill in the blanks.

1. A _ O _ P _ T _ R is a machine that can do many things.

2. A K _ Y _ O _ R _ is used to give instructions to a computer with the help of keys.

3. _ P _ A _ E _ S are used to listen to music and songs.

4. A _ O _ S _ is used to give instructions to the computer by pointing and clicking.

C Write True or False.

1. All the work we do on computer, is controlled by a CPU.

2. Printer displays the work done on the computer.

3. CPU is the most important part of a computer system.

4. A keyboard does not have number keys.

D Match the following computer parts.

1.



(a)

Speakers

2.



(b)

Printer

3.



(c)

Keyboard

4.



(d)

Monitor

5.



(e)

Mouse

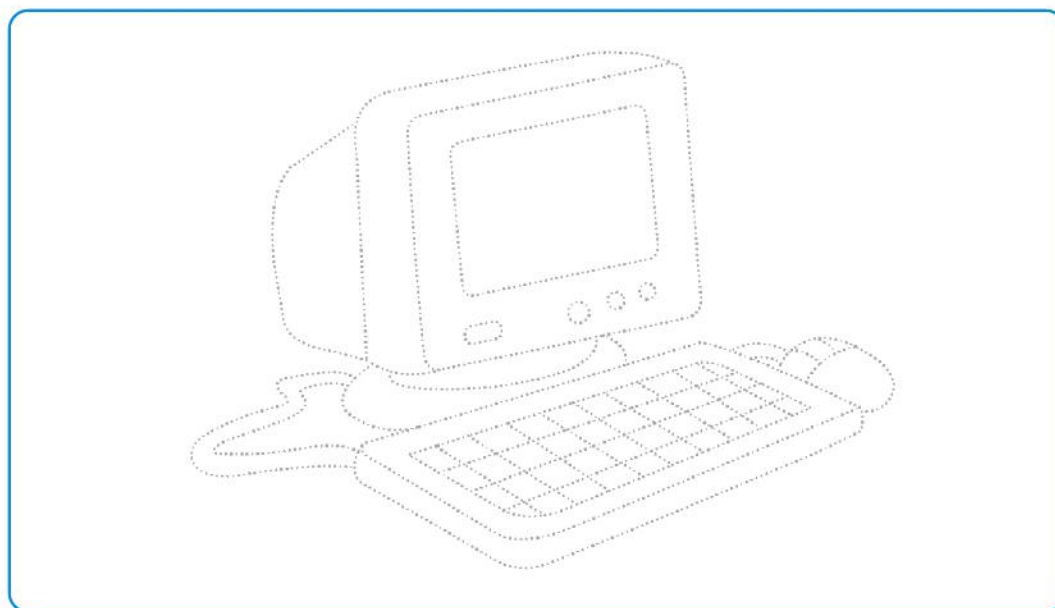


Refreshment Corner

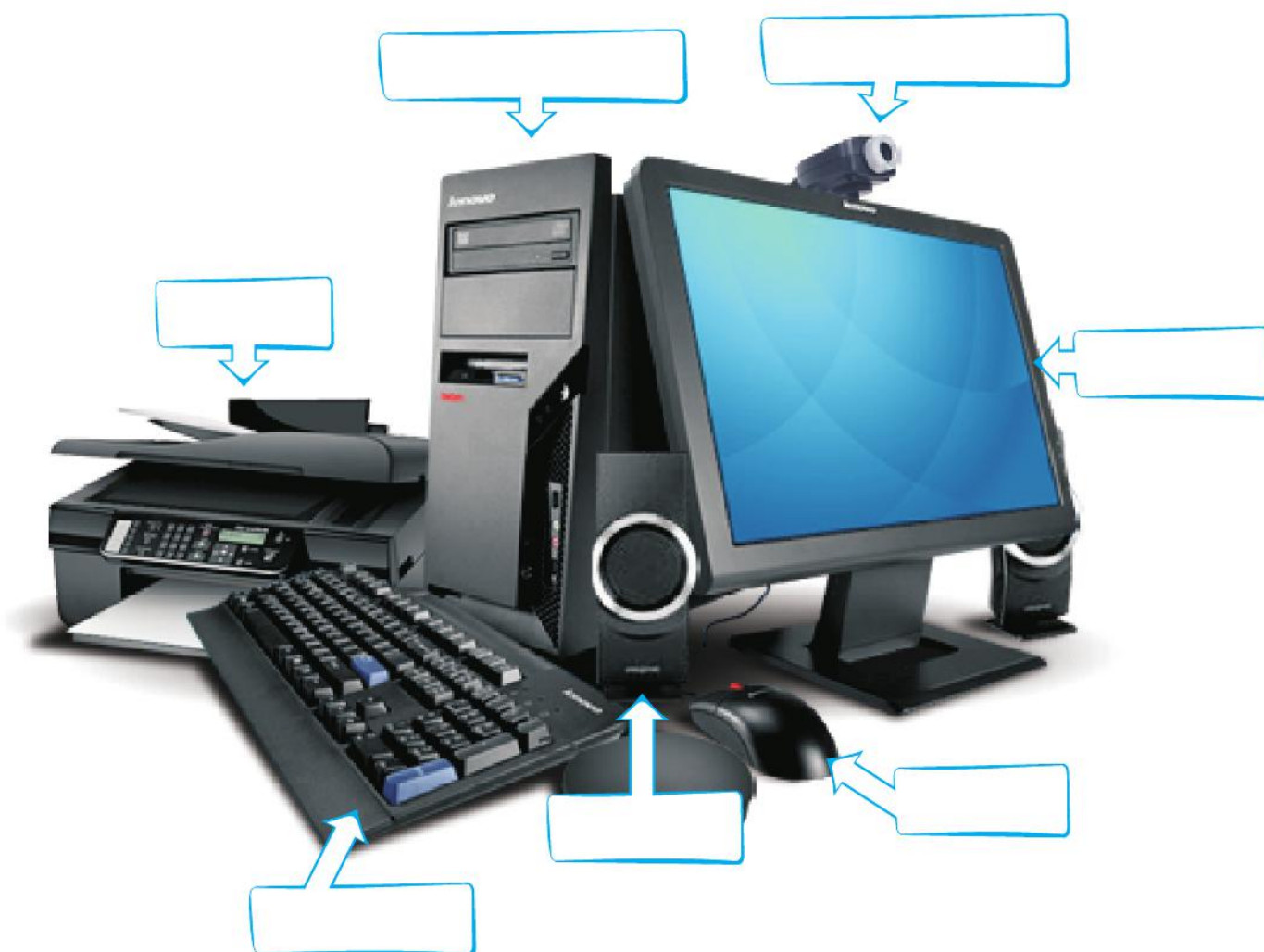


A Join the dots and colour the picture.

Art Integration



B Label the computer parts.



RACKING YOUR BRAIN

Life Skills and Values

One of your friends is not using a parts of the computer properly, then what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different parts of a computer in the class in detail.



TEACHER'S NOTE

Show the various parts of a computer and explain their functions to the students.



UNDERSTANDING THE MOUSE



Outlines

- Mouse
- Mouse Clicking

Mouse

A mouse is used to select the items on the monitor with an arrow.

When the mouse moves, the arrow also moves on the screen. The arrow is called the **mouse pointer**.



The mouse is kept on a flat surface called the **mouse pad**.



The mouse contains two buttons and one wheel. The buttons are used by clicking them and the wheel is used by scrolling.



Mouse is also called a 'Pointing Device'.



By clicking these buttons, we can tell the computer what to do. Wheel is used to move the page up and down.

Mouse Clicking

There are three types of mouse clicking:



Double click:

Pressing the left button twice, quickly. It is used to open or start any activity on the computer.



Left Click:

Pressing the left button once. It is used to select anything on the computer.

Right Click:

Pressing the right button once. It is used to control the activity after you have started it.



A wireless mouse gives you more freedom to move your mouse around.



Let's Implement.

A Tick (✓) the correct option.

1. A mouse is used to select the items on the monitor with an:

(a) arrow

☐

(b) button

☐

2. The arrow is called the mouse:

(a) pointer

☐

(b) tail

☐

3. The mouse is kept on a flat surface called the mouse:

(a) pad

☐

(b) wheel

☐

4. The buttons on mouse are used by:

(a) scrolling

☐

(b) clicking

☐

B Fill in the blanks.

1. The mouse contains two buttons and one __ H __ L.
2. Pressing the left button once, is called the __ E __ T click.
3. Pressing the left button twice quickly, is called the D __ U __ L __ click.
4. Pressing the right button once, is called the __ I __ T click.

C Write True or False.

1. When the mouse moves, the arrow also moves on the screen.
2. Left click is used to control the activity after you have started it.
3. Right click is used to select anything on the computer.
4. By clicking the buttons, we can tell the computer what to do.

D Colour the words that are related to the computer mouse.

Right click

Photo click

Mouse pad

Left click

Rat

Scroll wheel

Double click

Tail

E Match the columns.



1. Left click
2. Right click
3. Scrolling wheel
4. Double click



- (a) Move up and down a page
- (b) Start an activity
- (c) Select an activity
- (d) Control the activity



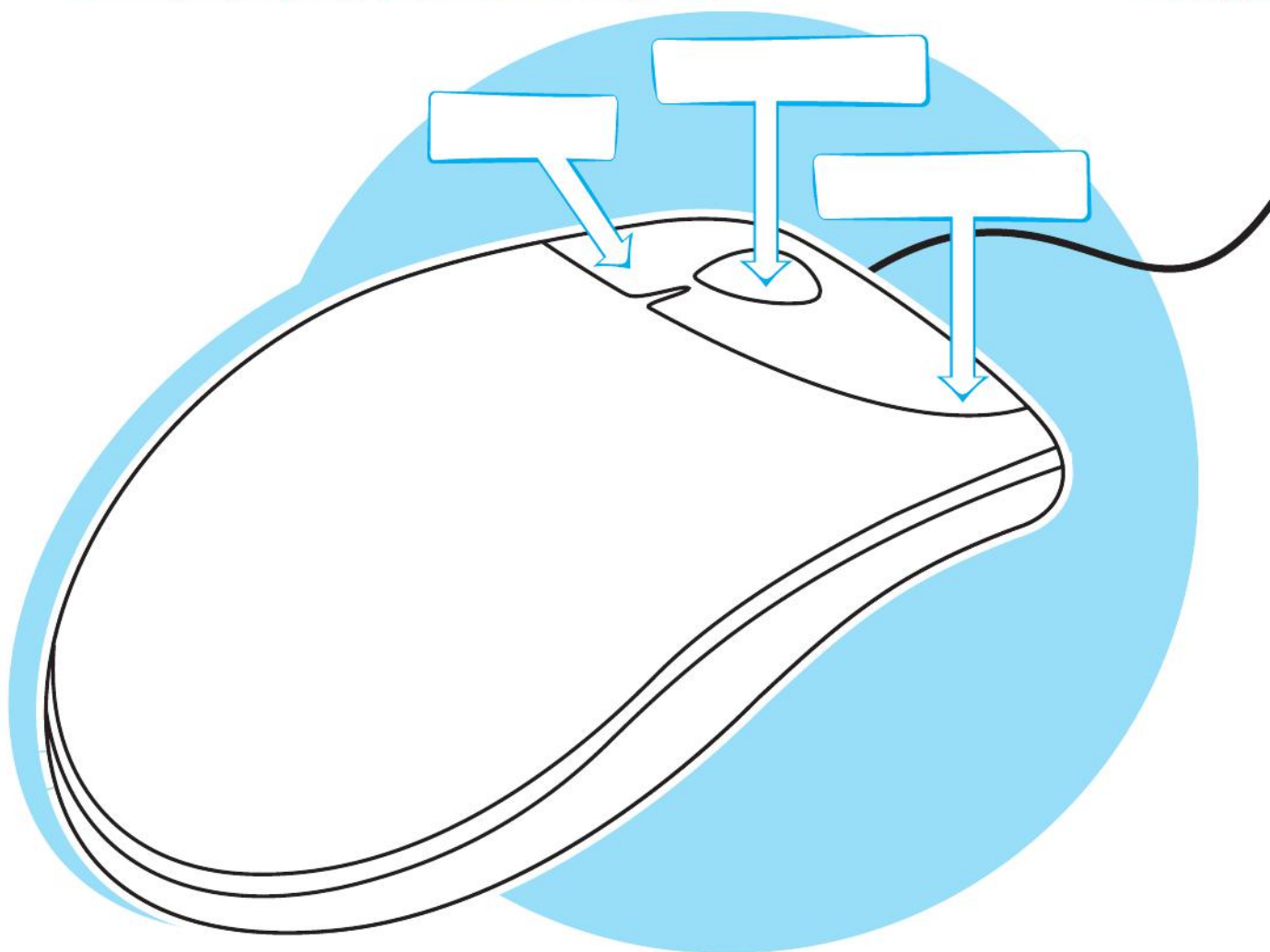
A Find the hidden words.

Mouse, select, arrow, screen, pointer,
buttons, wheel, click, scroll, pad

A	S	P	C	L	I	C	K	B
S	M	O	U	S	E	A	R	U
C	F	I	S	E	Y	O	O	T
R	P	N	D	L	H	W	W	T
O	A	T	F	E	I	H	A	O
L	D	E	S	C	R	E	E	N
L	O	R	H	T	I	E	D	S
A	A	R	R	O	W	L	E	R

- B** Colour the mouse with left button red, right button green, scroll button blue and the rest of the mouse yellow and label its parts.

Art Integration



RACKING YOUR BRAIN

Life Skills and Values

Do you play with the wire of a computer mouse or pull it? Is it a good or bad manner? Justify your answer.

Just Have a Discussion

Communication

Discuss the different mouse actions in the class in detail.



TEACHER'S NOTE

Demonstrate to the students how to hold and use a computer mouse.



QUICK REVIEW 1

Tick (✓) the correct option.

1. A very small computer that fits in your palm.
(a) Netbook ☐ (b) PDA ☐
2. Computer is used to keep records of patients:
(a) at the hospital ☐ (b) at the bank ☐
3. It is used to take the printout of our work.
(a) Speakers ☐ (b) Printer ☐
4. The buttons on mouse are used by:
(a) scrolling ☐ (b) clicking ☐
5. This computer will set up in a permanent location.
(a) Desktop ☐ (b) Laptop ☐
6. Computer is used to store a list of books and their locations.
(a) at the library ☐ (b) at the office ☐
7. A monitor is also called :
(a) LCU ☐ (b) VDU ☐
8. The mouse is kept on a flat surface called the mouse :
(a) pad ☐ (b) wheel ☐
9. This computer is a portable computer.
(a) Desktop ☐ (b) Laptop ☐
10. Computer is used to keep records of students and teachers:
(a) at the hospital ☐ (b) at the school ☐



THE KEYBOARD



Outlines

- Alphabet Keys
- Number Keys
- Special Keys

- ▶ A keyboard is an important part of a computer.
- ▶ A keyboard is connected by a wire to the CPU of the computer.
- ▶ We use a keyboard to give commands to the computer.
- ▶ A keyboard has many small buttons.
- ▶ The buttons on the keyboard are called **keys**.

There are different types of keys present on a keyboard. Let us learn about them.

Alphabet Keys

The keys you see from A to Z are called **alphabet keys**. They help us to write words.



Keyboard used by us is called 'QWERTY' keyboard.



Number Keys

The keys you see from 0 - 9 are called **number keys**. They help us to write numbers.



Special Keys

The keys on which you find words are called **special keys**. These keys are used for a specific purpose like copy paste.



Space bar

This key is used to give space between two letters.

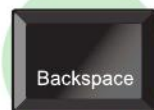
This key is the longest key on the keyboard.



Backspace

This key is like an eraser.

This key is used to delete the character from left of the cursor.



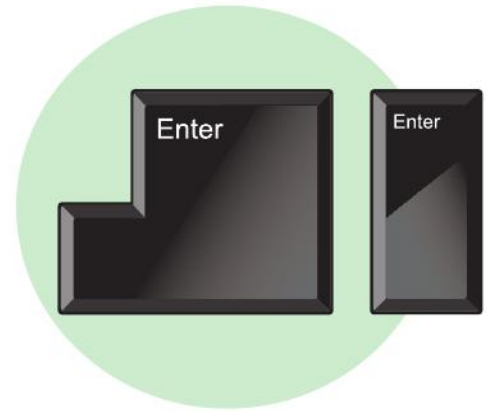
There are some keys on the keyboard which have two symbols on them.



Enter Key

This key is used to move to the next line.

There are two enter keys on the keyboard.



Let's Implement.

A Tick (✓) the correct option.

1. The buttons on the keyboard are called:

(a) Numbers

☐

(b) keys

☐

2. The keys on which you find words are called:

(a) Special keys

☐

(b) Number keys

☐

3. This key is used to give space between two letters.

(a) Backspace key

☐

(b) Spacebar key

☐

4. This key is used to delete the character from left of the cursor.

(a) Backspace key

☐

(b) Spacebar key

☐

B Fill in the blanks.

1. We use a keyboard to give ___ O ___ M ___ N ___ S to the computer.

2. The keys you see from A to Z are called ___ L ___ H ___ B ___ T keys.

3. The keys you see from 0 - 9 are called ___ U ___ B ___ R keys.

4. ___ P ___ C ___ A ___ keys are used for a specific purpose like copy paste.

C Write True or False.

1. A keyboard is connected by a wire to the CPU of the computer.
2. A keyboard has many small buttons.
3. Number keys help us to write words.
4. Space bar key is the longest key on the keyboard.

D Match the following.

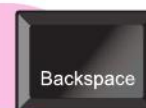
1. Alphabet keys

(a)



2. Number keys

(b)



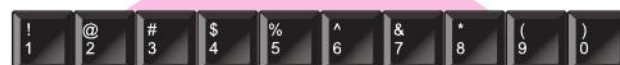
3. Special keys

(c)



4. Space bar

(d)



5. Backspace

(e)



6. Enter Key

(f)





Refreshment Corner



A Colour the alphabet keys that spell your name with yellow.



B Tick '✓' the correct way to sit while using the keyboard.



RACKING YOUR BRAIN

Life Skills and Values

What will you do if the keys of your keyboard stop working?

Just Have a Discussion

Communication

Discuss the uses of different keys of a keyboard in the class in detail.



TEACHER'S NOTE

Open a text editor and give simple exercises to the students so that they can get familiarised with the keyboard.



FUN WITH MS PAINT

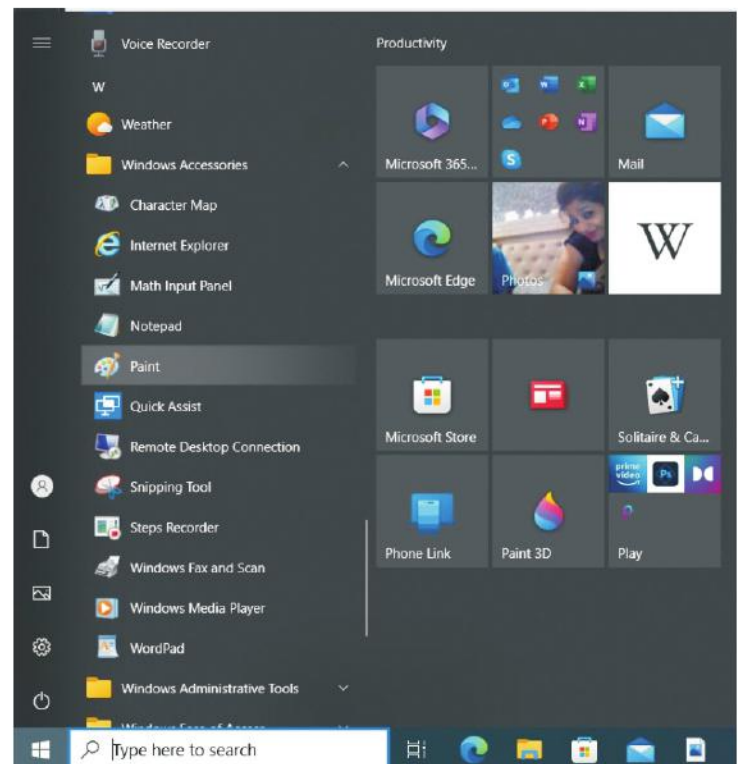


Outlines

- How to Start Paint?
- Brushes
- Tools in Paint
- Drawing Shapes

How to Start Paint?

1. Click on the **Start** button.
2. Select **All Apps**.
3. Choose the **Windows Accessories** folder.
4. Click on the **Paint**.

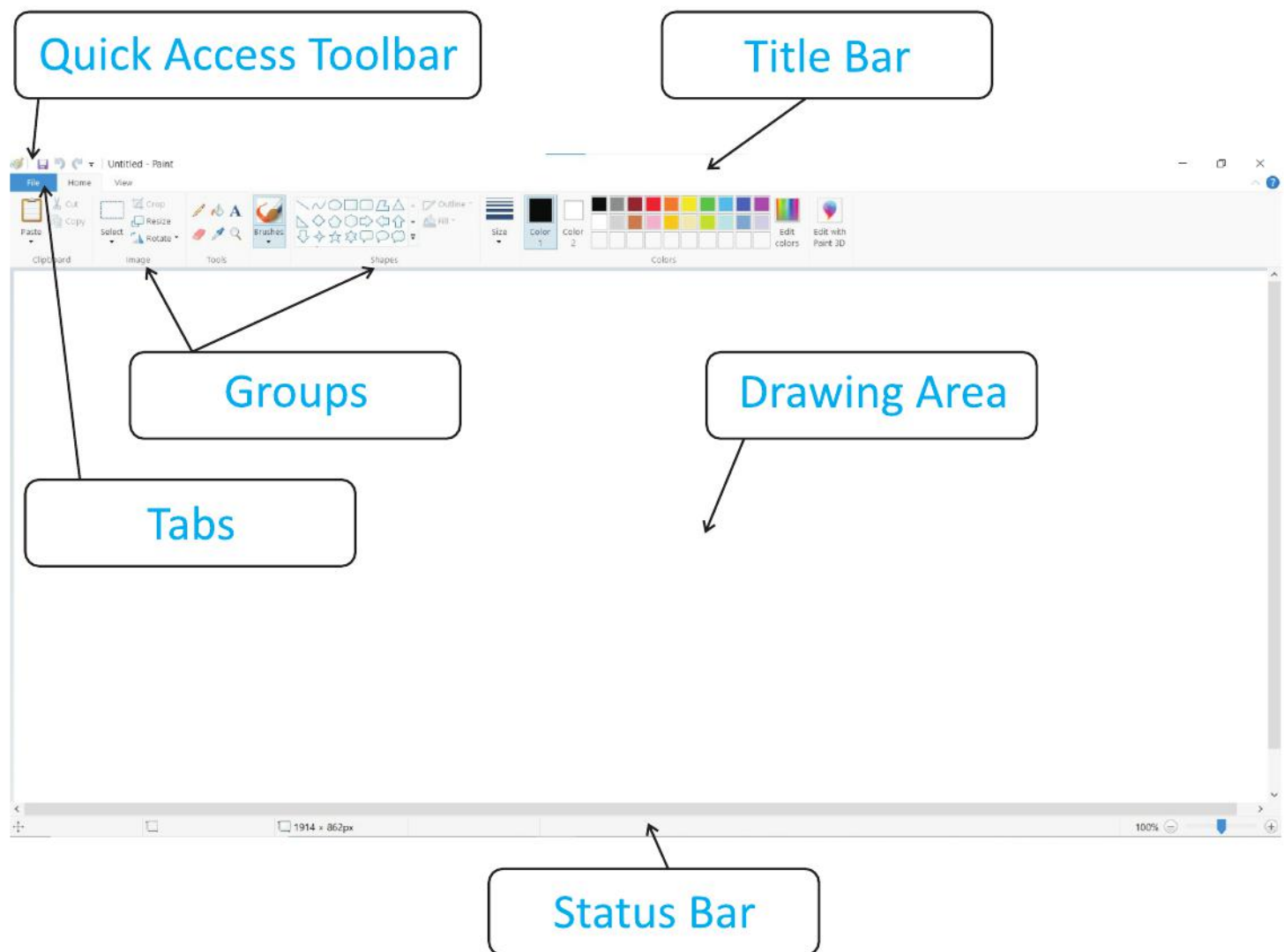


Smart Tip



You can also open **Paint** by typing **Paint** in the **Search Box** next to the **Start** button. Then, click on the **Paint** button.

The Paint window will appear as shown below:



The first version of **MS Paint** was introduced with the first version of windows in **1985**.

Tools in Paint

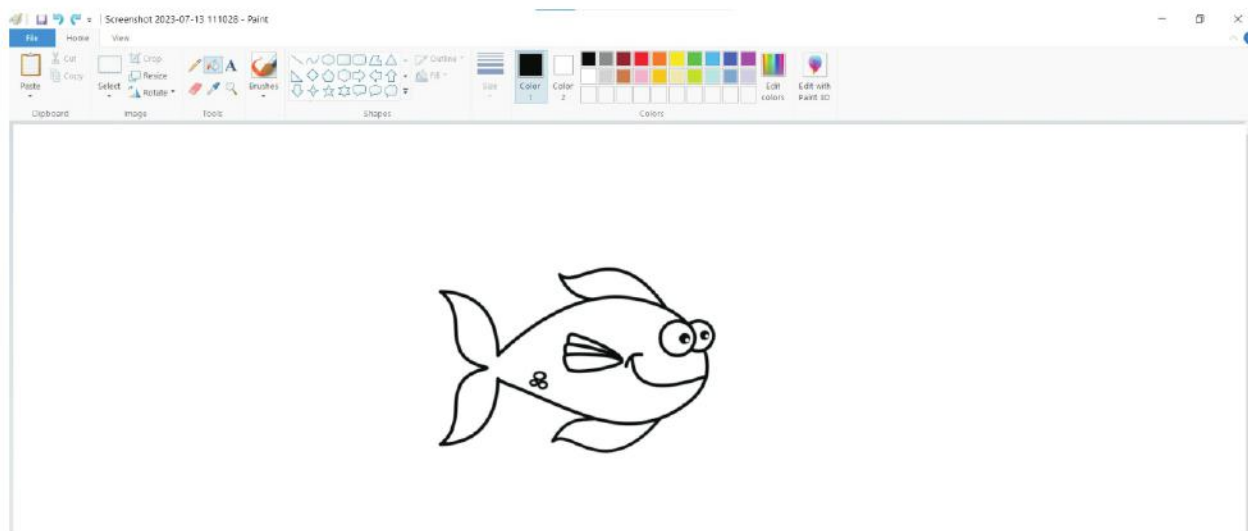
There are various tools in Paint to draw and colour. Let us now learn about the different tools in Paint.

Pencil Tool



The **Pencil** tool is used for freehand drawing. Follow these steps:

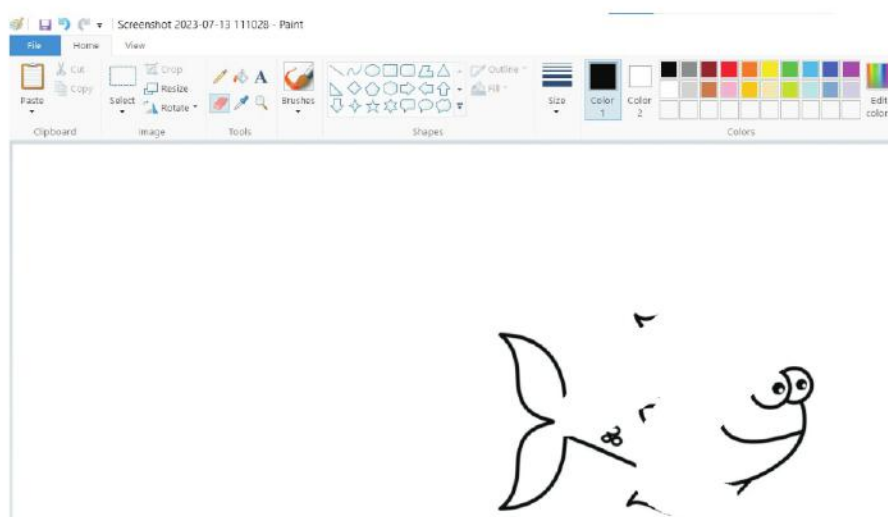
1. Click on the **Pencil** tool in the **Tools** group.
2. Click on the **Size** and select the thickness of the line.
3. Select the colour of your choice from the **Colors** group.
4. Now, press and hold down the left mouse button and drag the mouse pointer on the drawing area.



Eraser Tool

The **Eraser** tool can be used to erase any part of a picture. Follow these steps:

1. Click on the **Eraser** tool in the **Tools** group.
2. Click on the **Size** and select the Eraser size.
3. Now, press and hold down the left mouse button and move it on the area you want to erase.

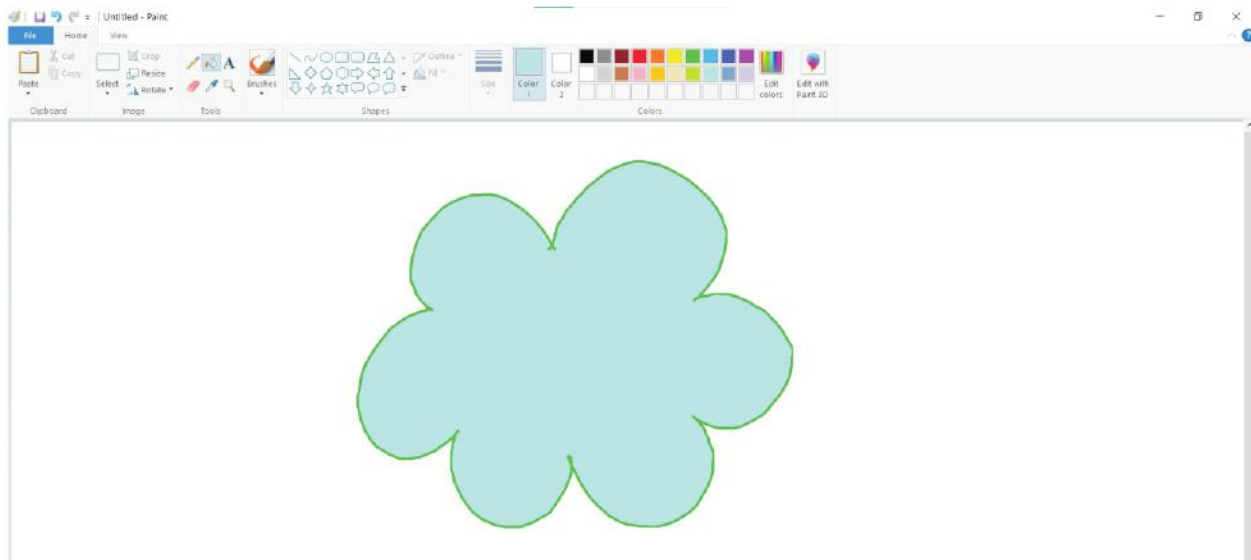


Fill with Color Tool



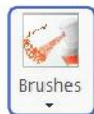
The **Fill with color** tool is used to fill colour in your drawing.

1. Click on the **Fill with color** tool in the **Tools** group.
2. Select the colour of your choice from the **Colors** group.
3. Left-click the mouse inside the area which you want to fill.



Note: You can use the **Fill with color** tool to fill colour in closed shape only.

Brushes

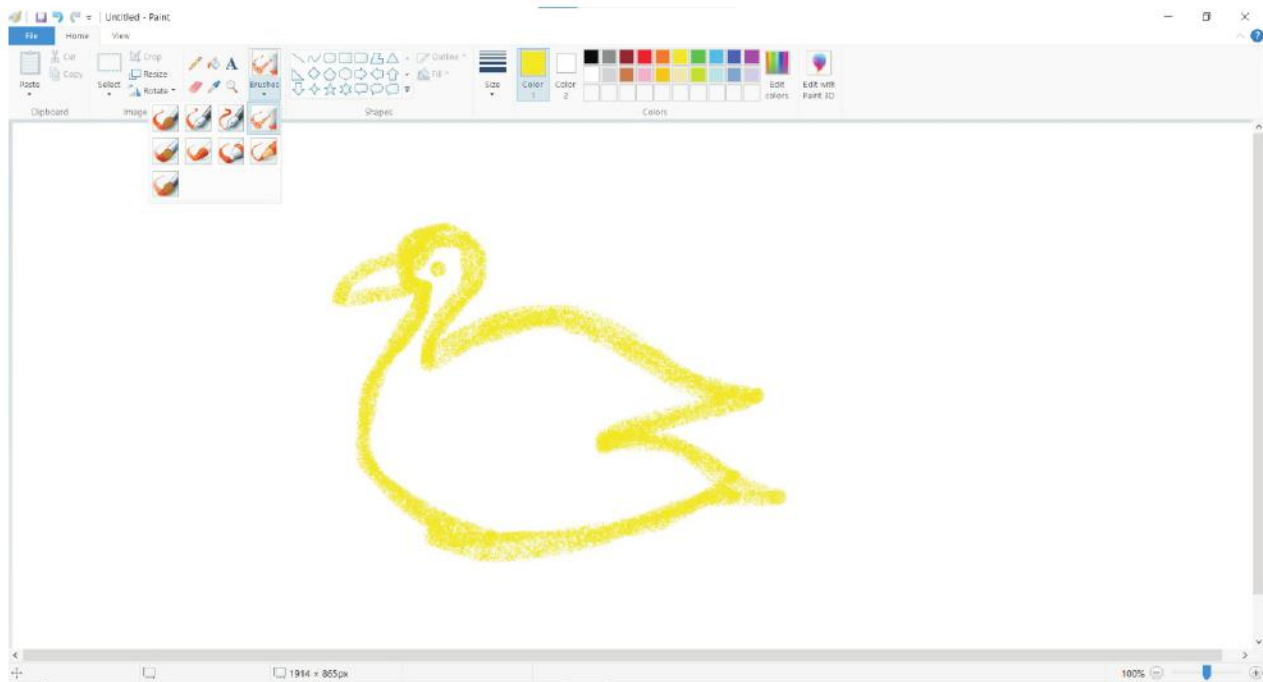


The **Brushes** tool is used to draw lines with different looks and effects.

Follow these steps :

1. Click on the **Brushes** on the **Home** tab.
2. Select the brush you want to choose.
3. Select the colour of your choice from the **Colors** group.

- Now, press and hold down the left mouse button and drag the mouse to draw.



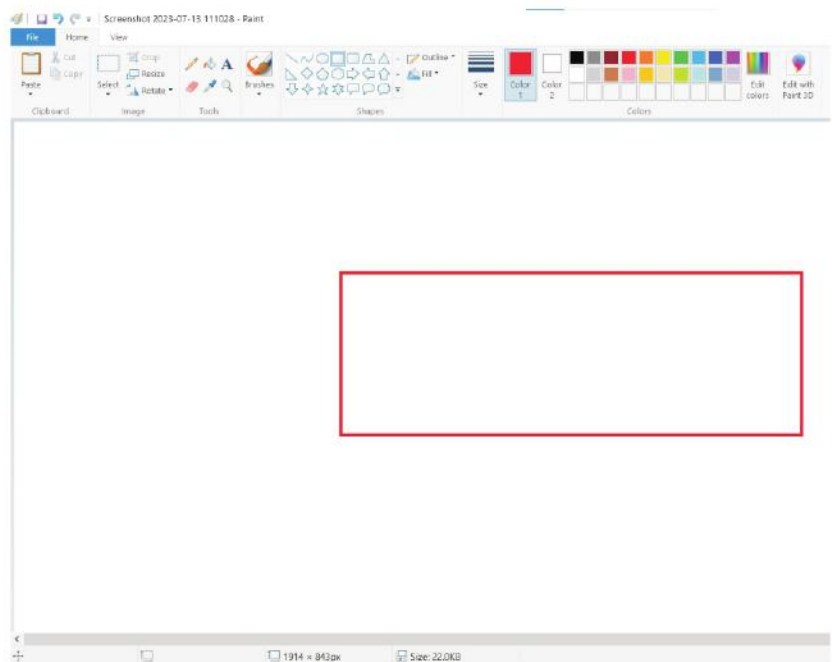
Drawing Shapes

Rectangle Tool



The **Rectangle** tool is used to draw a rectangle.

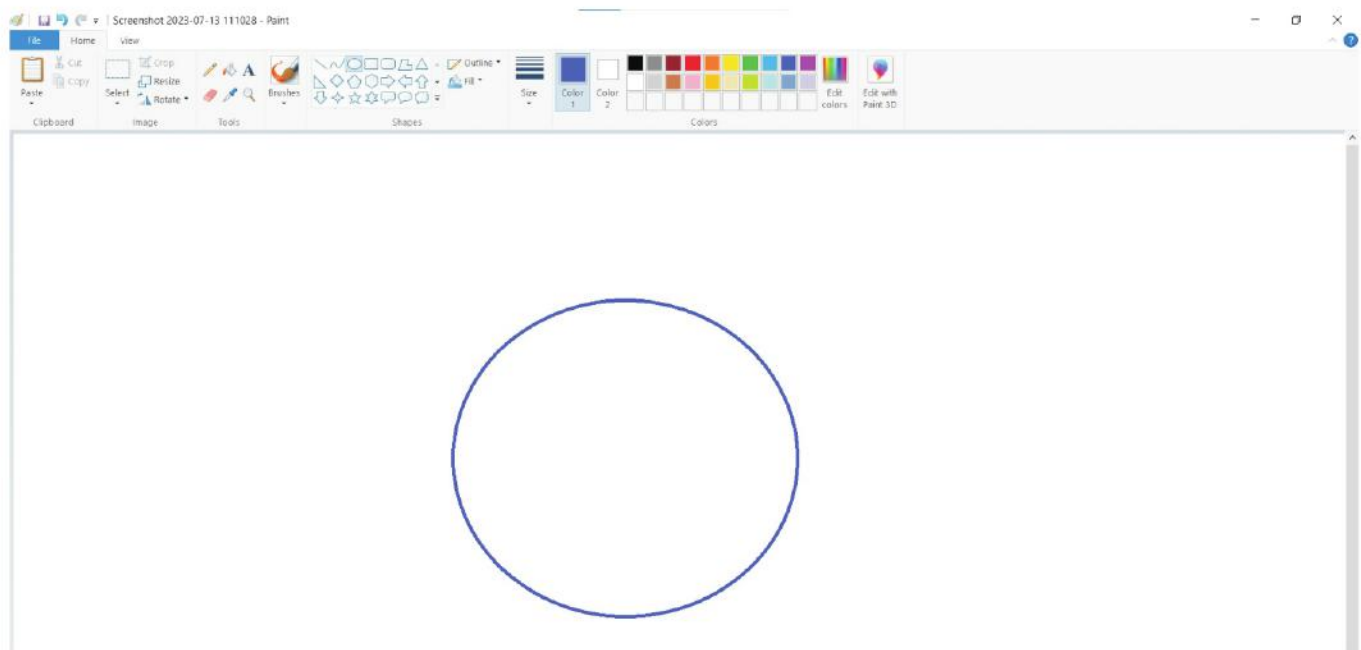
- Click on the **Rectangle** tool in the **Tools** group.
- Select the colour of your choice from the **Colors** group.
- Now, press and hold down the left mouse button and drag the mouse to draw a rectangle.



Oval Tool

The **Oval** tool is used to draw an oval shape or a circle. Follow these steps:

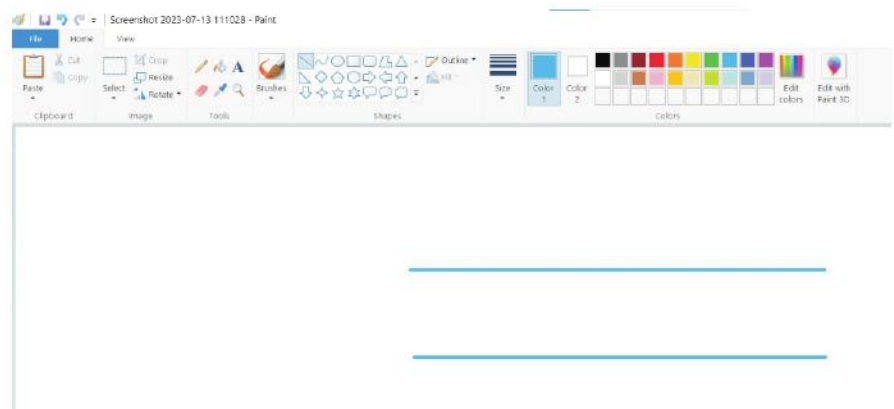
1. Click on the **Oval** tool in the **Tools** group.
2. Select the colour of your choice from the **Colors** group.
3. Now, press and hold down the left mouse button and drag the mouse to draw a circle.



Line Tool

The **Line** tool is used to draw a straight line.

1. Click on the **Line** tool in the **Tools** group.
2. Select the colour of your choice from the **Colors** group.



- Now, press and hold down the left mouse button and drag the mouse to draw a line.



Smart Tip

To draw a straight line, press and hold down the

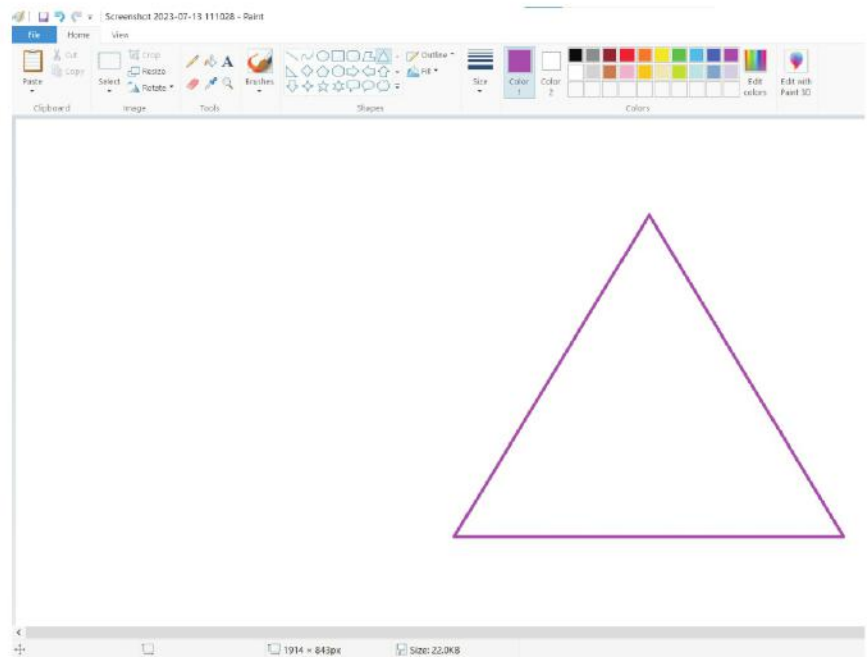
↑ Shift

Key while dragging the mouse on the drawing area.

Triangle Tool

The **Triangle** tool is used to draw a triangle.

- Click on the **Triangle** tool in the **Tools** group.
- Select the colour of your choice from the **Colors** group.
- Now, press and hold down the left mouse button and drag the mouse to draw a triangle.



Let's Implement.

A Tick (✓) the correct option.

- The _____ tool is used for freehand drawing.

(a) Pencil



(b) Eraser



- The Eraser tool is present in the _____ group.

(a) Size



(b) Tools



3. The _____ tool is used to draw an oval shape or a circle.

(a) Circle



(b) Oval



4. The _____ tool is used to draw lines with different looks and effects.

(a) Pencil



(b) Brushes



B Fill in the blanks.

1. The Pencil tool is used for _____ drawing.

2. The _____ tool is used to draw a rectangle.

3. The _____ tool can be used to erase any part of a picture.

4. You can draw a straight line using the _____ tool.

C Write down the steps to start the Paint.

D Match the following.



(a) is used to draw a rectangle.



(b) is used for freehand drawing.



(c) is used to erase any part of a picture.



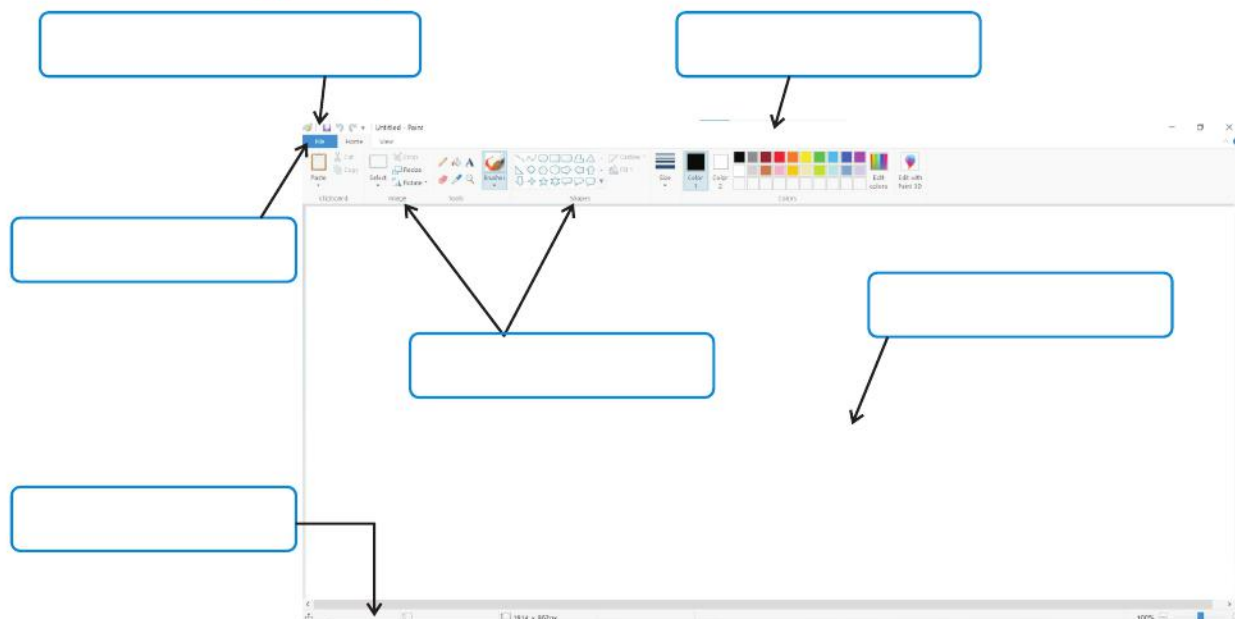
(d) is used to fill colour in your drawing.



(e) is used to draw a straight line.

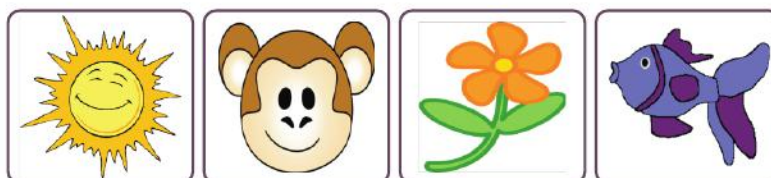
Refreshment Corner

- Label the figure given below.



Experiential Learning

- Visit your computer lab and open Paint. Draw and colour the following.



RACKING YOUR BRAIN

Life Skills and Values

One of your friends is not able to work with Paint, then what will you do? Justify your answer.

Just Have a Discussion

Communication

Discuss the different tools of Paint in class in detail.



TEACHER'S NOTE

- Give enough exercises to students so that they get practice of using the tools of Paint as well as using the mouse.



WHAT IS ARTIFICIAL INTELLIGENCE?



Outlines

- Drones
- AI and Our World
- What is Artificial Intelligence?

Drones

The various uses of drones are as follows:

- ▶ Drones can be used to deliver medicines to faraway places.
- ▶ Drones can be used by farmers for watering crops.
- ▶ Drones can take the pictures and videos at various heights.
- ▶ Drones are also being used to deliver goods and foods.



This is all possible with the use of **Artificial Intelligence (AI)**.

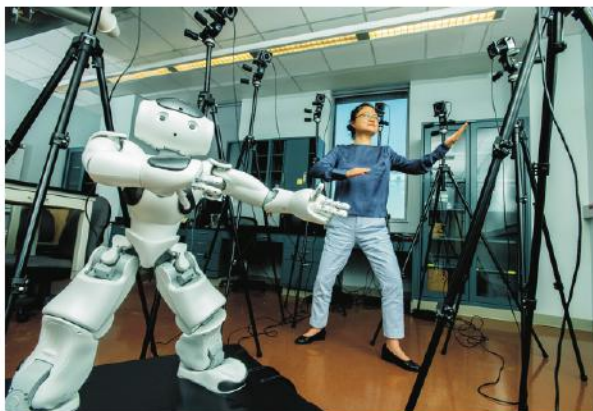
What is Artificial Intelligence?

AI is the ability that helps machines or robots to do tasks on

their own, much like humans. AI-enabled machines do the defined work on their own. They are controlled by the computer.

Some of the machines based on AI are as follows:

► Robots



► Alexa, etc.



AI and Our World

Artificial Intelligence has become an important part of our lives. We are using some sort of AI in many devices. Let us take a look at some of these devices.

Amazon Echo

The Amazon Echo is a voice-controllable speaker. It is an AI-based smart speaker that can interact with you. It comes with a built-in virtual assistant named Alexa that can answer your queries, follow your instructions, and control your smart home devices. You can ask Alexa to play your favourite music or audiobooks, set alarms, provide



weather forecasts and make calls to the contacts on your smartphone.

Self-driving Cars



Driverless or self-driving car or an autonomous car is a vehicle that navigates between locations by using a variety of sensors, cameras, radar, and AI. Without assistance from humans, it travels over roads

to reach a designated location. Passengers can spend time doing leisure activities during a trip as there is no need to drive in self-driving cars. Audi, BMW, Ford, Google, General Motors, Tesla, Volkswagen and Volvo are some of the most well-known companies developing or testing autonomous cars.

Smartphones

These are portable devices which combine the functions of a mobile phone and computing devices. The most obvious example of AI on cell phones is in the voice assistant. No matter what family your smartphone is from, if you have an assistant like Siri or Google Assistant, there is Artificial Intelligence. Voice recognition and its synthesis are the clearest examples of this technology. Smartphones use AI for a variety of tasks such as face recognition, voice assistants, navigation, weather status, photo filter, games, etc.



Smartwatches

Smartwatches are small smartphone-like gadgets worn on the wrist that provide a variety of features that traditional watches don't have. It is connected to a smartphone that notifies the user of incoming calls, email messages and notifications from applications. You can even make calls using a smartwatch. They are used to detect a patient's diabetic condition, blood pressure, heartrate, count your steps and other regular health issues.



Chatbot



Chatbot is an AI application used to make an online chat conversation. It uses AI to understand customer's questions and automate responses to them, simulating human conversation.

These are very commonly used by companies like amazon, Flipkart, etc. for customer support.

Smart TVs

A smart TV does not require a set-top or other streaming device. It is connected to the internet for streaming apps or channels, like Netflix, Amazon Prime, Hotstar, etc.



A Smart TV comes with an in-built virtual assistant, like Google or Alexa, using which you can operate your TV. For example, you can ask Alexa to turn on or off your TV, play your favourite apps or shows, and so on. It shows you a list of programmes or movies based on your preferences. You can even play games and do web browsing using a smart TV.

Smart Doorbells

Smart doorbells are an AI enabled and inform the home owner when a visitor arrives at the door. A smart doorbell is an internet-connected doorbell that notifies the smartphone or other electronic device of the home owner when a visitor arrives at the door. It activates when the visitor presses the button of the doorbell, or alternatively, when the doorbell senses a visitor with its built-in motion sensors.



Let's Implement.

A Tick (✓) the correct option.

- _____ is an AI application used to make an online chat conversation.
(a) E-mail ☐ (b) Chatbot ☐
- Smartphones use AI for _____.
(a) face recognition ☐ (b) voice assistants ☐

3. _____ can take the pictures and videos at various heights.

(a) Alexa

☐

(b) Drones

☐

4. _____ is/are based on AI.

(a) Humans

☐

(b) Robots

☐

B Fill in the blanks.

1. _____ can be used by farmers for watering crops.

2. _____ machines do the defined work on their own.

3. _____ has become an important part of our lives.

4. _____ or _____ car or an autonomous car is a vehicle that navigates between locations.

C Write True or False.

1. The most obvious example of AI on cell phones is in the voice assistant.

2. Smart doorbells do not inform the home owner when a visitor arrives at the door.

3. A Smart TV does not come with an in-built virtual assistant.

4. AI-enabled machines are controlled by the computer.

D Write down the various uses of drones.



Refreshment Corner



- Draw and colour the picture of Alexa in the space given below.

Art Integration

RACKING YOUR BRAIN

Life Skills and Values

Smart devices are used to make our tasks easier. Should we always use them at small tasks? Justify your answer.

Just Have a Discussion

Communication

Discuss the different devices in which AI is used in class in detail.



TEACHER'S NOTE

Explain the use of different drones and other smart devices to students in detail.



CODE WITH SCRATCHJR



Outlines

- What is ScratchJr?
- Starting ScratchJr
- Changing the Background
- Saving a Project
- Features of ScratchJr
- Creating a New Project
- Changing a Character

What is ScratchJr?

ScratchJr is a computer program or app used to create animated stories and games. In ScratchJr, blocks are joined to make characters move, jump, dance and sing. Let us learn about ScratchJr in detail.

Features of ScratchJr

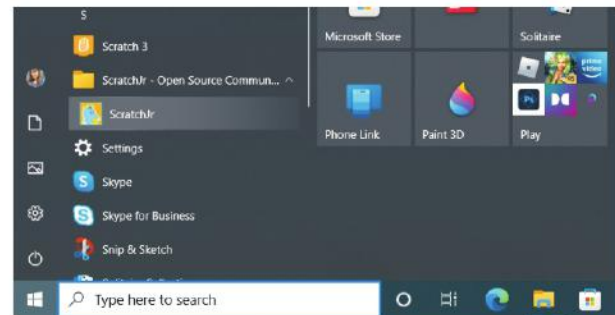
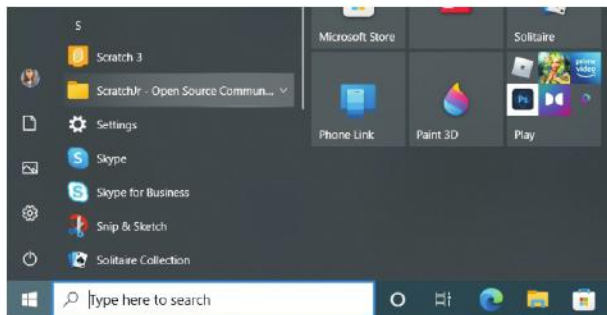
Features of ScratchJr are as follows:

- ▶ Easy to learn coding
- ▶ Simple coding blocks

Starting ScratchJr

To start ScratchJr, follow these steps:

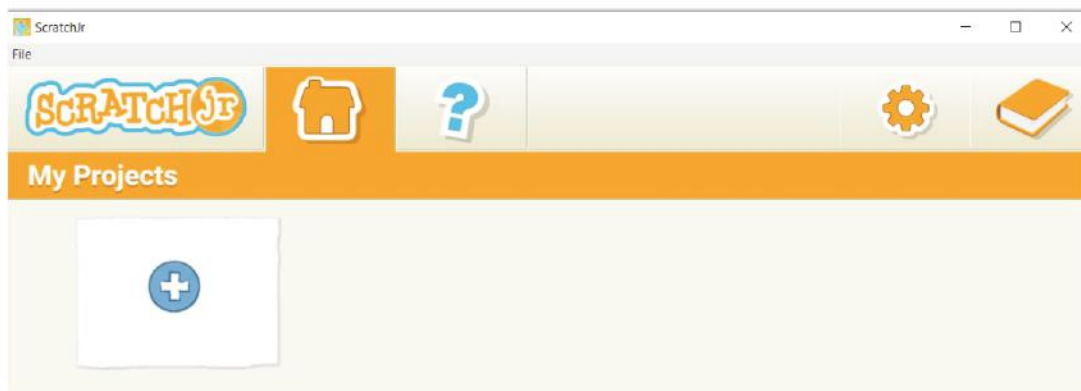
1. Click on the **Start** button.
2. Click on the **ScratchJr- Open Source Communication** folder.
3. Select **ScratchJr**.



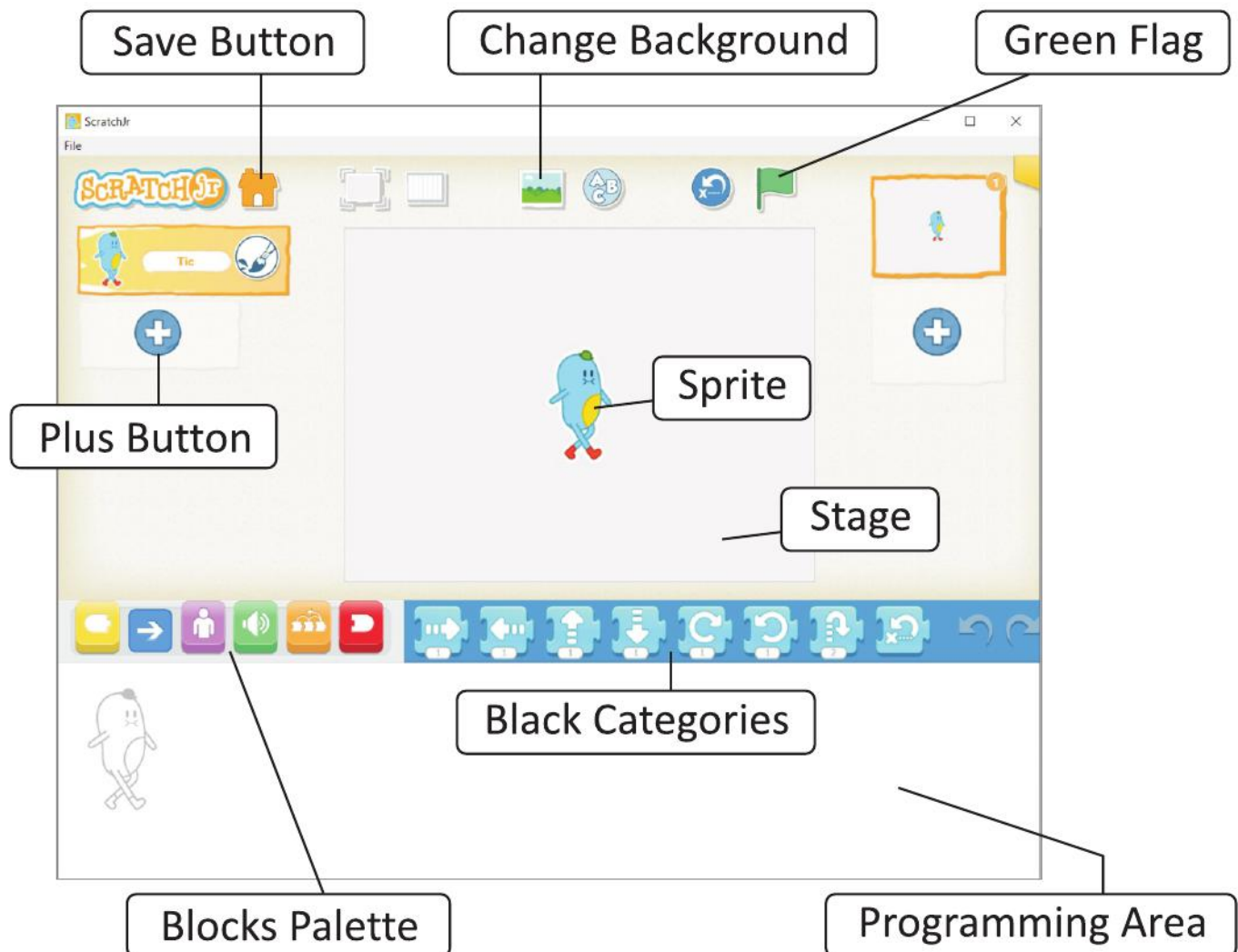
The screen given below appears.



1. Click on the **Home** button. The screen given below appears.



2. Now, click on the **Plus** sign to start a new project. The first window of ScratchJr will appear as given below.



The various components of ScratchJr are as follows:

Sprite

It is the main character that performs action.

Stage

It is the main working area where characters move and perform actions.

Change Background

It is used to select a background for the stage.

Green Flag

It is used to run a Scratch project.

Blocks Palette

It displays different types of blocks based on their function and use.

Reset Character Button

It resets all the characters to their starting position on the stage.

Save Button

It is used to save the project and return to the Home page.

Plus Button

It is used to add characters.

Block Categories

It is the menu of programming blocks. It includes blocks like: Triggering (Yellow), Motion (Blue), Looks (Purple), Sounds (Green), Control (Orange) and End (Red).

Programming Area

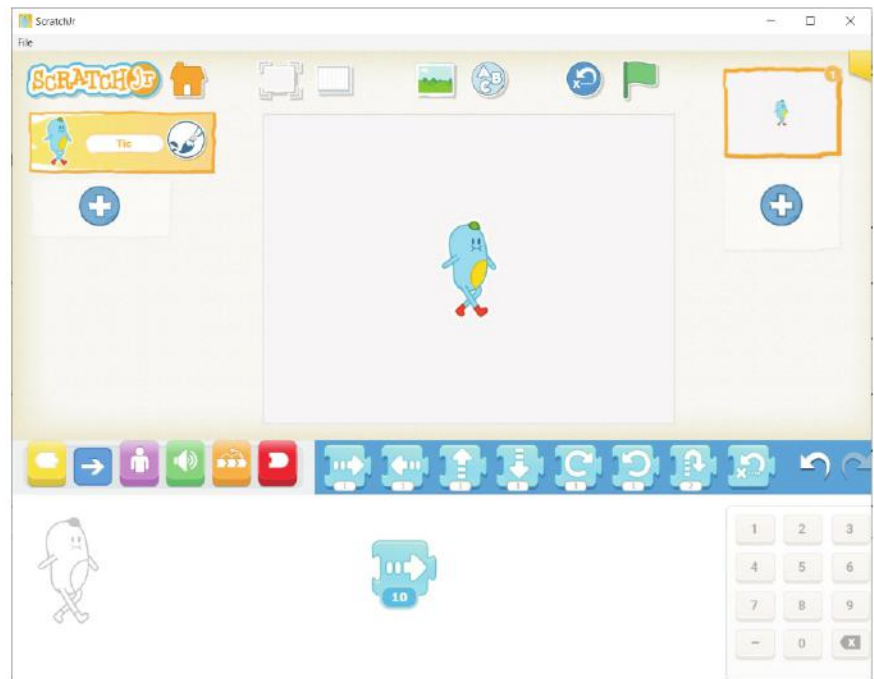
It is where you connect programming blocks to create scripts.

Creating a New Project

To create a new project, follow these steps:

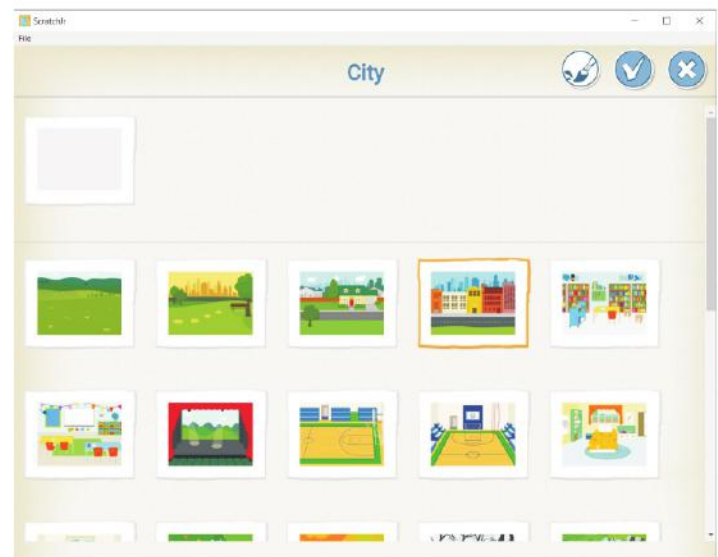
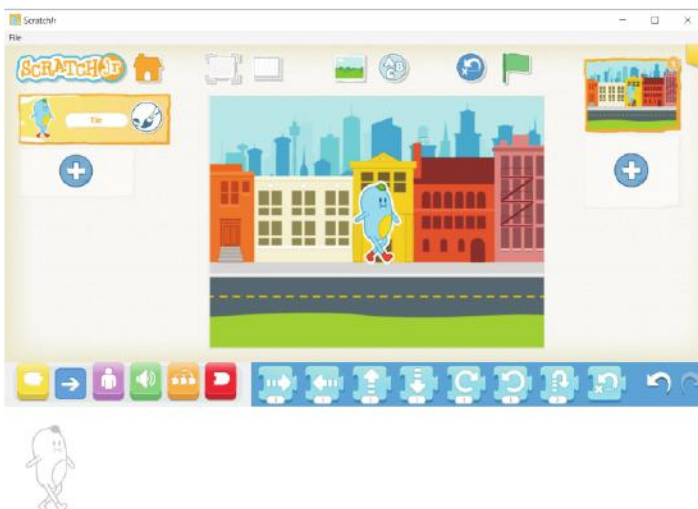
1. Drag the **MOVE RIGHT** block in the coding area.
2. Click on the number written in the **MOVE RIGHT** block.

3. Change the value to 10 by using the number grid at the right side of the stage.
4. Click on the block to see the sprite moving at the right side of the stage.



Changing the Background

1. Click on the **Background** button  to add a new background. The screen given below appears.
2. Select the background of your choice.

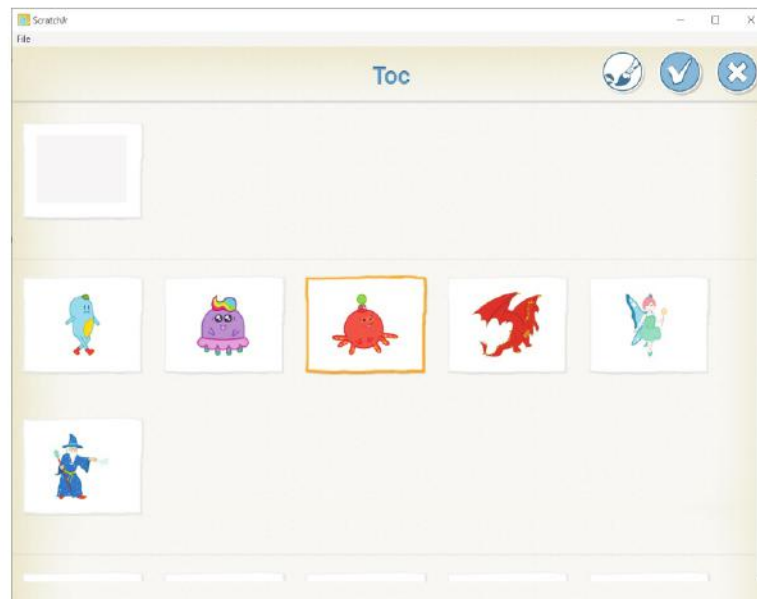


3. Click on the **Tick** sign to add the selected background.

Changing a Character

To change a character, follow these steps:

1. Bring the pointer on the default sprite. Press the left mouse button and hold it. A red cross sign appears on the sprite. Click on it to remove the default sprite.
2. To add a new sprite, click on the **Plus** sign in the left pane. The screen given below appears.



3. Select the selected child and click on the **Tick** sign to add it.



Saving a Project

To save the project, click on the Save button. ScratchJr automatically saves the file with a number added to the word 'Project' like project1, project2, and so on.



Let's Implement.

A Tick (✓) the correct option.

- _____ is the main character that performs action.
(a) Sprite ☐ (b) Stage ☐
- _____ Flag is used to run a Scratch project.
(a) Home ☐ (b) Green ☐
- _____ palette displays different types of blocks based on their function and use.
(a) Stage ☐ (b) Blocks ☐

B Fill in the blanks.

- _____ is the main working area where characters move and perform actions.
- _____ is used to select a background for the stage.
- _____ is used to save the project and return to the Home page.
- _____ is used to add characters.

C Write True or False.

- ScratchJr is used to create animated stories and gamers.

2. Home is the main working area.
3. Blocks palette displays different types of blocks.
4. The new characters can be added from the Backgrounds Library.

D Identify the following pictures and write their names.







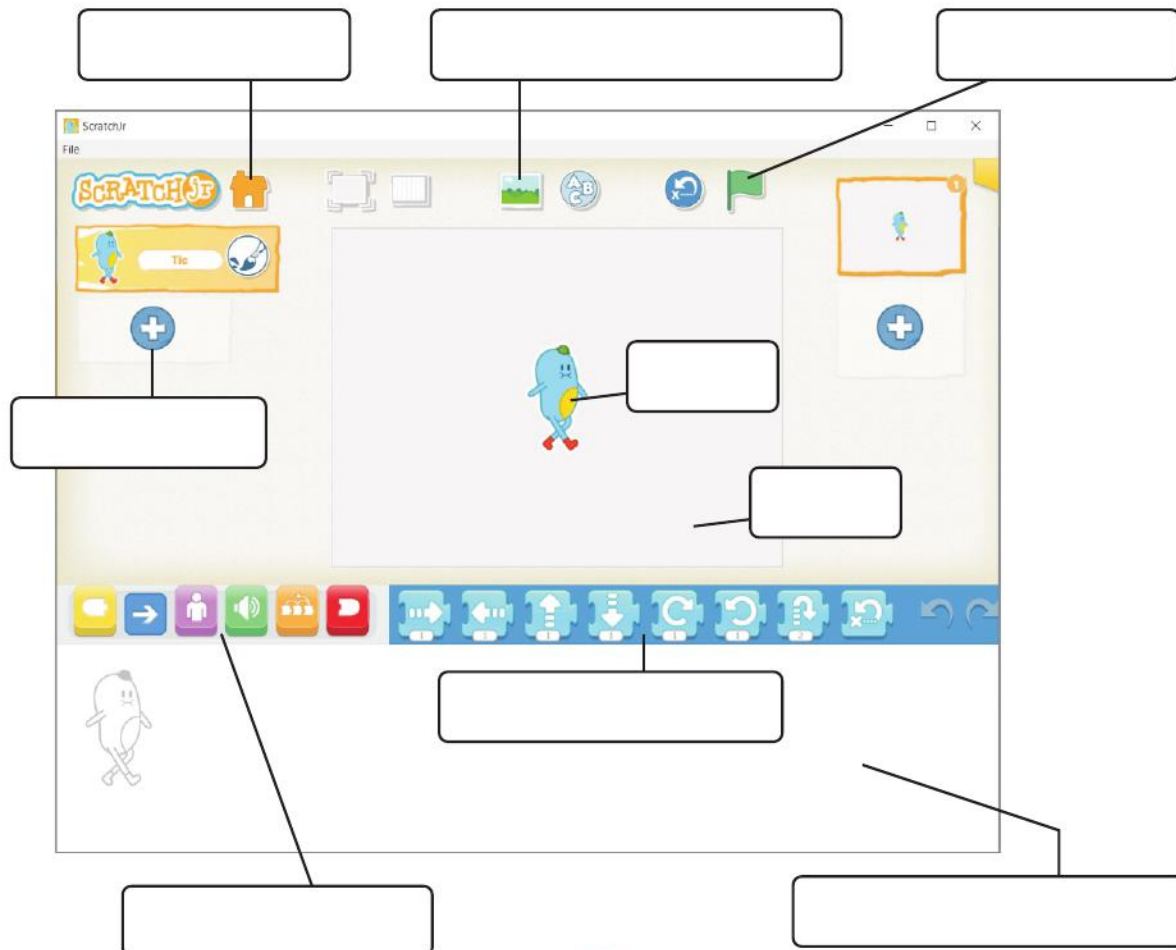




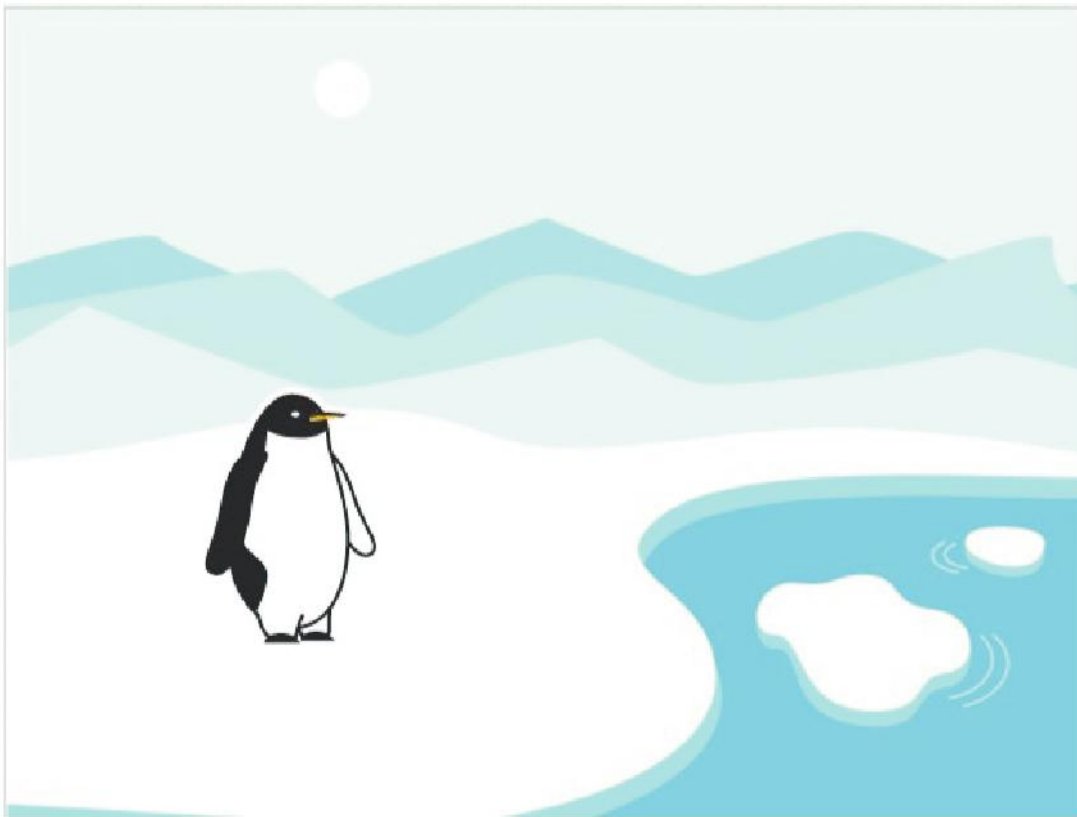
Refreshment Corner



► **Label the figure give below.**



- Perform the following tasks in ScratchJr given below.



RACKING YOUR BRAIN

Life Skills and Values

Your friend is not able to add a new background in ScratchJr, then what will you do? Will you help your friend or not? Justify your answer.

Just Have a Discussion

Communication

Discuss the different components of ScratchJr in the class in detail.



TEACHER'S NOTE

Demonstrate the different blocks to the class in detail.



QUICK REVIEW 2

Tick (✓) the correct option.

1. The buttons on the keyboard are called:
(a) Numbers ☐ (b) Keys ☐
2. The Eraser tool is present in the _____ group.
(a) Size ☐ (b) Tools ☐
3. _____ can take the pictures and videos at various heights.
(a) Alexa ☐ (b) Drones ☐
4. _____ Flag is used to run a Scratch project.
(a) Home ☐ (b) Green ☐
5. The keys on which you find words are called:
(a) Special keys ☐ (b) Number keys ☐
6. The _____ tool is used to draw an oval shape or a circle.
(a) Circle ☐ (b) Oval ☐
7. _____ is/are based on AI.
(a) Humans ☐ (b) Robots ☐
8. _____ is the main character that performs action.
(a) Sprite ☐ (b) Stage ☐
9. This key is used to give space between two letters.
(a) Backspace key ☐ (b) Spacebar key ☐
10. The _____ tool is used for freehand drawing.
(a) Pencil ☐ (b) Eraser ☐



A Fill in the blanks.

1. Pressing the left button once is called __ E __ T click.
2. Computer helps in playing the G ____ E ____.
3. The mouse contains two buttons and one __ H ____ L.

B Write True or False.

1. Computer does not work on our instructions.
2. Computer does not help in writing the letters.
3. By clicking the buttons, we can tell the computer what to do.
4. Left click is used to control the activity after you have started it.

C Complete the words given below.

1. O P T R
2. E E T O I
3. D U L
4. E E T I I Y

D Colour the words that are related to the computer mouse.

Right click

Mouse pad

Left click

Photo click

Rat

Scroll wheel

Tail

Double click



A Fill in the blanks.

1. We use a keyboard to give __ O __ M __ N __ S to the computer.
2. You can draw a straight line using the _____ tool.
3. _____ has become an important part of our lives.
4. _____ is used to select a background for the stage.

B Write True or False.

1. A keyboard is connected by a wire to the CPU of the computer.
2. The new characters can be added from the Backgrounds Library.
3. Smart doorbells do not inform the home owner when a visitor arrives at the door.

C Match the following.

- | | |
|--|---|
| 1.  | (a) is used to draw a rectangle. |
| 2.  | (b) is used for freehand drawing. |
| 3.  | (c) is used to erase any part of a picture. |
| 4.  | (d) is used to fill colour in your drawing. |
| 5.  | (e) is used to draw a straight line. |